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The Late Carboniferous to Early Triassic geodynamic evolution of Variscan Europe in the light of magnetic overprints in Early Permian rhyolites from the northern Vosges (France) and central Black Forest (Germany)

J. B. Edel[1] and J. L. Schneider[2]

[1] *Institut de Physique du Globe, 5 rue Descartes, 67084 Strasbourg Cedex, France*
[2] *Université des Sciences et Techniques, Laboratoire de Pétrologie, Bat.* SN5, 59655, *Villeneuve d'Ascq Cedex, France*

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SUMMARY

This paper demonstrates how magnetic overprints of a geological series can provide information over a long time period, which can be interpreted in terms of geotectonic evolution. According to these new results, the Late Carboniferous-Early Permian rhyolites from the northern Vosges have recorded the magnetic field over a major part of the Permian. Recent radiometric dating assigns a Late Carboniferous–Early Permian age (298 Ma) to the previously 'Middle Permian' rhyolitic volcanism of the northern Vosges. A palaeomagnetic study was undertaken on rhyolites of the Nideck–Donon massif and on neighbouring Permian basins (Villé, Saint-Dié) in order to obtain a new pole for the Early Permian and to detect overprints. Standard palaeomagnetic and rock magnetic experiments demonstrate that over 90 per cent of the characteristic directions of magnetizations are of secondary origin. The high-temperature magnetizations carried by secondary haematite and haematized titano-magnetite fail the fold test, and display directions consistent with the Saxonian–Thuringian directions from Variscan Europe (Nideck-Donon: 186°/−17°, $k = 167$ for 26 sites, VGP: 50° N/178° E; Saint-Dié basin: 184°/−20.5° for two sites, VGP: 53° N/180° E; Villé basin: 191.5°/−25° for two sites, VGP: 53° N/168° E; Vosges dikes: 197.5°/−26.5° for two sites, VGP: 53° N/158° E). Part of the magnetizations with intermediate unblocking temperatures of magnetite consists also of post-tectonic overprints. Two consistent mean directions with lower inclinations, obtained in the Nideck–Donon rhyolites (182°/−7°, $k = 459$ for 13 sites, VGP: 45° N/184° E) and pyroclastic deposits from the Villé basin (178°/−9°, $k = 519$ for three sites, VGP: 46° N/190° E), were probably acquired during Late Autunian-Early Saxonian times. In rhyolites the pre- and post-tectonic characteristic magnetizations are carried by magnetite, while in the ash-fall deposits the carrier is allogenic haematite. A few older magnetizations (191/−3, two sites, VGP: 42° N/173° E), consistent with magnetizations carried by magnetite from rhyolites of the central Black Forest, and results from coeval volcanics as well as from Autunian sediments have been observed in tuffs from the Villé basin (188°/1.5°, two sites, VGP: 40° N/177° E. The oldest magnetizations (207°/10°, $k = 329$, VGP: 32° N/155° E), presumably of Stephanian age, were identified in two rhyolitic stocks within the granitic basement from the central Vosges and central Black Forest. The distribution of the palaeomagnetic directions suggests that, from the Late Autunian to the Late Thuringian, overprinting due to low-temperature alteration of titano-magnetite and crystallization of secondary haematite was more or less continuous.

The apparent polar wander path (APWP) computed with the new results and the published poles shows a hairpin, which implies a drastic change of the European plate motion during the Permian. The clockwise rotation of Europe initiated in the Late Visean–Namurian stopped in the Late Autunian–Early Saxonian. This event corresponds to the end of the Variscan convergence and of the Appalachian

orogeny. In the northern Vosges, the hinge of the APWP is also associated with the tectonic phase responsible for the tilting of the volcanic layers. The motion of the European plate was then converted into a counterclockwise rotation and a northward drift until the Late Triassic.

Key words: magnetic overprints, Northern Vosges, Permian, Polar wander path, rhyolites.

INTRODUCTION

During Late Carboniferous–Early Permian times, Europe was affected by crustal thinning in the aftermath of the Carboniferous collision of the Variscan orogeny (Lorenz & Nicholls 1984). Intermontane troughs with clastic sedimentation and extensive volcanic activity of calcalkaline affinity characterize this extensional period, which ceased in the late Early Permian. It was followed by a period of peneplanation of the Variscan Belt. In the Late Permian, regional subsidence led to transgression of the Zechstein Sea in the north, and of the Tethys Ocean in the south. Generally, the Permian appears to be a tectonically quiet period. This is contested by Prost & Becq-Giraudon (1989), who document a mid-Permian compressive tectonic event in Europe, contemporaneous with the late Alleghanian phase of the Appalachian orogeny. In southern and central Europe, the change of the tectonic regime occurred in the Latest Autunian and is demonstrated by the Saxonian unconformity and by folding (Broutin *et al.* 1994).

The Permian volcanics and sediments of western Europe have been the subject of extensive palaeomagnetic studies in the last three decades (Van Everdingen 1960; Nijenhuis 1961; Roche, Saucier & Lacaze 1962; Kruseman 1962; Birkenmaier, Krs & Nairn 1968; Krs 1968; Westphal 1972; Konrad & Nairn 1972; Berthold, Nairn & Negendank 1975; Zijderveld 1975; Mauritsch & Rother 1983; Courtillot *et al.* 1986; Evans & Maillol 1986; Merabet & Daly 1986; Thompson, Guillaume & Daly 1986; Westphal *et al.* 1987; Soffel & Harzer 1991; Merabet & Guillaume 1988; Cogné, Brun & Van den Driessche 1990; Diego-Orozco & Henry 1992). The resulting mean palaeopoles, for the various areas studied, scatter significantly. This is partly due to poorly separated magnetizations, but mostly to inconclusive fold tests as demonstrated in the Lodève basin (Cogné *et al.* 1990). In the volcanics of the Intra-Sudetic basin, the dispersion was explained by the coexistence of primary and secondary directions, the primary having shallow inclinations and more or less rotated declinations and the secondary showing higher inclinations, consistent with the expected Late Permian direction (Westphal *et al.* 1987). Previous investigations on the Permian Nideck–Donon rhyolites in the northern Vosges also show a scatter in inclination, interpreted as reflecting two separate volcanic phases, in Stephanian–Autunian and Saxonian–Thuringian times (Roche *et al.* 1962; Westphal 1972). Thus, despite the great number of data, the Permian polar wander curve for Europe is still poorly defined. The reasons seem to be the widespread occurrence of overprints, leading to unclear ages of magnetizations, and to uncertainties concerning the tectonic attitude of secondary directions.

The aims of this new work in the Nideck–Donon units were:

(1) to obtain new radiometric and palaeomagnetic data for the volcanic rocks of the northern Vosges;

(2) to establish whether Permian volcanic rocks have been remagnetized;

(3) to examine other data published for Varican Europe in the light of these new results;

(4) on the basis of all data, to check whether a prominant geotectonic event has occurred during the Permian.

This palaeomagnetic investigation has been carried out together with a volcanological study which aimed to unravel the mechanisms of emplacement of the volcanic series. To obtain complementary structural information (foliation, flow direction, etc.), samples were taken from a large area for anisotropic magnetic susceptibility as well as for palaeomagnetic measurements. Palaeomagnetic results presented here include also some results from sites without clear bedding criteria.

GEOLOGICAL SETTING

The Late Carboniferous–Permian units of the northern Vosges

In the northern Vosges, the Late Carboniferous–Permian formations unconformably overlie the Devonian–Viséan basement and crop out in fault-bounded depressions. Farther to the north, they are unconformably covered and hidden by Triassic sedimentary deposits ('Grès Vosgien'). Generally, the Permian formations are gently tilted north-westwards (0°–25°). The dip of the Triassic sedimentary cover is also north-westwards, but does not exceed 5°.

Volcanic rocks are represented by thick tuff and lava sequences of rhyolitic composition (Saucier, Millet & Jost 1959a, b; Siat 1974; Benzakour 1988). Microgranitic and granitic intrusions as well as minor andesitic and trachyandesitic lava flows are associated with the rhyolites (Rocci & Chrétien 1963; Benzakour 1988). Alluvial and lacustrine deposits are locally interbedded with the volcanic layers. The volcanic activity is recorded in three zones of the northern Vosges (Fig. 1).

The Nideck–Donon zone. This zone represents the main volume of exposed volcanic rocks (Mihara 1935; Saucier *et al.* 1959a, b; Benzakour 1988), and consists of rhyolitic pyroclastic breccias, ignimbrites, and lava flows. These facies are associated with clastic deposits and are overlapped by aeolian deposits of Thuringian age (Hollinger 1978). In the upper Sarre Blanche valley, a small microgranitic body intrudes the lower pyroclastic units.

The Saint-Dié basin. This is the most important Permian formation in the Vosges (Choubert & Gardet 1935; Vollrath

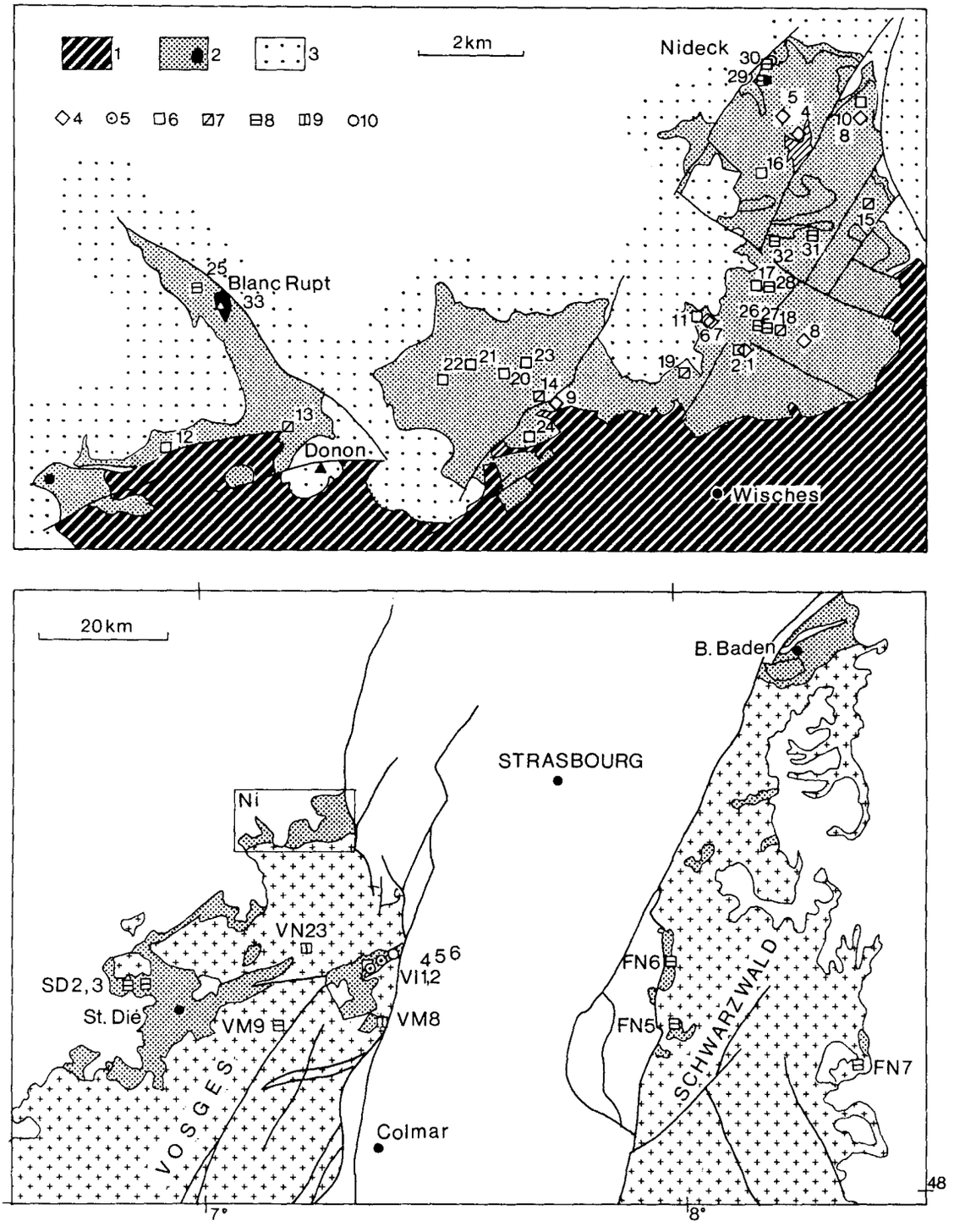


Figure 1. Geological sketch map of the Permian units and location of palaeomagnetic sites (NE France and SW Germany). (1) Devonian–Devonian basement; (2) Permian volcanites and microgranite; (3) Triassic cover; (4) rhyolitic tuffs and breccias; (5) ash-tuffs; (6) ignimbrites; (7) lava-like ignimbrites; (8) rhyolitic flows; (9) acidic dikes; (10) sediments.

1945; Hollinger 1970). Clastic rocks represent the main part of the deposits of this intermontane basin. Rhyolitic and trachy-andesitic volcanic rocks ('Meisenbückel' layers) are limited to the bottom of the unit and crop out in the Saint-Michel–Nompatelize area (Fig. 1).

The Villé basin. The Villé basin is a small intermontane pull-apart basin (Laubacher & von Eller 1966; Carasco 1987, 1989). Conglomerates, sandstones and lacustrine mudstones as well as limestones dominate the sedimentary inventory. Rhyolitic pyroclastic rocks are occasionally interbedded.

Rock types and sampling

The main sampling has been conducted in the northern Vosges (Nideck–Donon zone, Saint-Dié basin and Villé basin). In addition to the layered Permian units, an acidic dike (VM 8) and a Late Carboniferous–Early Permian rhyolitic stock within the granitic basement were also sampled (VM 9) (Fig. 1). For discussion, we will also consider the result from another acidic dike of the northern Vosges (VN 23) published in a previous paper (Edel *et al.* 1986) as well as results from radiometrically dated rhyolites of the central Black Forest (FN 5, 6, 7; Edel 1987) (Fig. 1).

Intrusive rocks

Microgranite (rhyolite de Blanc Rupt, Site Ni 33). Microgranite is exposed in a dome-shaped circular body and intrudes rhyolitic flows (Siat 1974) (Fig. 1).

Volcanic rocks

Volcanic rocks consist mostly of rhyolites and minor andesites and trachy-andesites. Rhyolites show different facies, mainly pyroclastic, but also effusive.

Welded rhyolitic breccia (Sites Ni 1, 3, 4, 5, 6, 7, 8, 9) are only present at the bottom of the Nideck–Donon units. It is a coarse breccia with a large proportion of millimetric to decimetric rhyolitic lava and pumice clasts and some pelitic fragments. The fine matrix is vitroclastic and contains abundant euhedral embayed quartz and altered felspars. Pumice clasts are often flattened and lay stress on a rough fluidal texture. This pyroclastic breccia seems to result from violent explosive eruptions.

Rhyolitic lappili and ash tuffs (Sites Vi 1–2, 4, 5) form thin graded sequences with an abundant silicified vitroclastic matrix containing euhedral and broken quartz, feldspar crystals and rare lithics. Only present in the villé basin ('Meisenbückel' series), lapilli and ash tuffs are distal equivalents of the rhyolitic breccias from Nideck–Donon.

Rhyolitic welded ignimbrites (Sites Ni 2, 10, 11, 12, 17, 18, 19, 20, 21, 22, 23, 24, Vi3). Ignimbritic columnar-jointed deposits occur in the Nideck–Donon zone (Saucier *et al.* 1959a, b; Siat 1974), where they form cliffs in the landscape. This facies is crystal-rich and displays a fluidal texture often shown by fiamme parallel to the palaeohorizontal. Ignimbritic deposits are discontinuous and seem to correspond to valley deposits (high-aspect ratio ignimbrites according to Walker 1983).

Rhyolitic lava-like ignimbrites (according to Eckren *et al.* 1984; Sites Ni 13, 14, 15). This facies overlies the ignimbrites of the Nideck–Donon zone. It consists of metric to decimetric and columnar-jointed depositional sequences. Deposits sometimes show an agglomerate aspect and have characteristics intermediate between the underlying ignimbrites and the overlying lava flows. The matrix displays a lamination parallel to the bedding and contains decimetric flattened juvenile clasts. This facies can be compared with the lava-like ash-flow tuffs described by Eckren *et al.* (1984) in Idaho, and seems to result from lava fountaining related to felsic fissural eruptions.

Lavas flows (Sites Ni 25, 26, 27, 28, 29, 30, 31, 32, SD 2, 3) occur in the Nideck–Donon zone, where they were previously interpreted as welded tuffs (Saucier *et al.* 1959a, b), and in the Saint-Dié basin, where they are found at the bottom of the series. The rhyolites form thick layers, finely laminated and without pyroclastic structures. This flow foliation often displays folds similar to those described in felsic lava flows (Fink 1983; Bonnischen & Kaufmann, 1987).

Sedimentary facies

Conglomerates and arkoses (Site Vi 6). These facies are the main deposits in the Saint-Dié and Villé basins. These siliciclastic deposits contain a high proportion of lithic material that attests to an intense erosion of the Vosges basement. Conglomerates and arkoses were deposited in alluvial fans and fluvial environments.

Tuffaceous breccias and sandstones. Clastics are rich in volcanic fragments and form lenticular deposits. These facies result from the erosion of the volcanic material and were emplaced by fluvial processes.

Aeolian sandstones (Grès de Champenay). The youngest deposits of the Permian deposits in the Vosges are arkosic sandstones characterized by well-sorted, rounded grains with tangential high-angle cross bedding. These characteristics are typical of dune facies emplaced in an aeolian environment (Hollinger 1978). Three sites were sampled but did not provide results that could be interpreted

Alteration of the magmatic rocks

In magmatic rocks, clay minerals, sericite or minor calcite often replace feldspars. Mafic minerals such as amphiboles are completely altered. However, biotites remain well preserved, hence it is possible to date them by radiometric processing (Montigny, in Schneider *et al.* 1994). Oxides consist mainly of haematite and titano-haemitite associated with rutile, but relics of titano-magnetite and ilmenite are also present. The glassy groundmasses of the effusive and pyroclastic rocks are completely devitrified. Felsitic and spherolitic textures are common. Volcanic glass of the groundmass, pumice and glass shards are replaced by cryptocrystalline aggregates of silica and alkali feldspars or by chlorite and calcite. All these observations strongly indicate that the volcanics underwent hydrothermal alteration during and after cooling.

Structural control

Bedding geometries were established in the pyroclastic deposits by measuring azimuth and dip (1) of the layer contacts, (2) of layering in interbedded volcanic-sedimentary units, (3) of fiamme plane and (4) of the axes of columnar joints. The axes of the columnar joints are perpendicular to the fiamme and to the upper cooling surface which, in ignimbrites and lava-like ignimbrites, are both parallel to the palaeohorizontal. Difficulties in finding the palaeohorizontal were only encountered in the lava flows. In some areas, the upper rhyolitic lava flows show a subvertical fan-like foliation alone E–W features which are interpreted as the

upper part of alimentation dikes (Ni31, 32). On both sides and close to the source, the foliation was strongly folded during emplacement of the flows. Farther away, folds disappear and the rhyolites become finely laminated with subhorizontal foliation. In the latter case, the lamination was considered to be close to the palaeohorizontal. In most sites, the anisotropy of magnetic susceptibility (AMS) provided complementary information on bedding (Schneider *et al.* 1994).

After emplacement, the volcanic and volcano-sedimentary units of the Nideck–Donon zone were tilted towards the NNW to WNW with angles of up to 20°. According to field observations and AMS measurements, tilting of the microgranite Ni 33 is negligible. An erosion surface separates the Late Permian Champenay sandstones from the Early Permian volcanics. The dip of the Triassic sandstone layers, which is deduced from contour levels, is only 3° towards the NW. The Late Permian sandstones are also subhorizontal. Consequently, a tectonic tilting phase affected the volcanic units between the Early and the Late Permian. This tectonic activity may be associated with the Saxonian discordance in the Lorraine basin.

Ages

Fossils only occur in the Villé basin. Plants and spores characteristic of Stephanian A were found in the lowest units and Autunian plants in the 'Assises d'Albé'. The Saxonian age attributed to the 'Meisenbückel' series, the 'Frapelle' series and the rhyolites is based on the stratigraphic position of these series, which overlie Autunian series. As there is sometimes a coexistence of Stephanian and Autunian floras, depending on climatic characteristics, in Permo–Carboniferous continental formations of Europe (Broutin *et al.* 1986), precise biostratigraphic datings are not available. To solve this problem, radiometric ages are needed.

Radiometric ages are available for rhyolites of the Nideck–Donon series. In the Nideck area, Lippolt & Hess (1983) obtained two ages, of 298.4 ± 6.7 and 299.5 ± 6.5 Ma, using the K/Ar method, and 294 ± 0.9 Ma by ^{40}Ar–^{39}Ar plateau ages. In the parallel volcanic study, K/Ar dating of biotite from the lava-like ignimbrite Ni 14 and the microgranite Ni 33 yielded respective ages of 297 ± 9 and 299 ± 9 Ma (Montigny in Schneider *et al.* 1994) which are consistent with the results from Lippolt & Hess (1983). The Carboniferous–Permian boundary being fixed at 290 Ma by Harland *et al.* (1989), 295 Ma by Odin (1994) and 300 Ma by Hess & Lippolt (1986), the radiometric ages of the rhyolitic magmatic rocks invalidate the Saxonian age inferred from stratigraphical criteria and locate the rhyolitic magmatism around the Carboniferous–Permian boundary. Using the Rb–Sr method on biotites, Lippolt, Schleicher & Roczek (1983) obtained similar ages, of 286 ± 5 Ma, on sites FN 5 and FN 7, respectively, in the central Black Forest.

RESULTS

Palaeomagnetic and rock-magnetic experiments

33 sites were used in the Nideck–Donon rhyolites, two sites in the Saint-Dié basin, five sites in the volcanics of the Villé basin, three sites in dikes and rhyolites within the Palaeozoic basement of the central Vosges and three sites in the central Black Forest. Five to ten samples were generally cored in one site, and each core was cut into two or three specimens. the direction and intensity of natural remanent magnetization was measured on a modified Digico spinner magnetometer. Susceptibility and anisotropy of magnetic susceptibility measurements were also performed on Digico devices in the framework of an investigation of the rhyolitic magmatism. IRM analysis, and thermal demagnetization of three-component IRM (Lowrie 1990), as well as examination of polished sections, were performed in order to identify the magnetic phases. 382 specimens were thermally demagnetized using home-made furnaces. Characteristic directions of remanence are obtained from line fitting Zijderveld diagrams with at least three points per line (usually five) (Kirschvink 1980).

The NRM of rhyolites and tuffs can be described by almost univectorial magnetizations (Fig. 2). Unblocking temperatures are generally high (650–680 °C). In some sites (Ni 3, 9, 10, 14, 15, 33, Vi 3, FN 6), components with intermediate temperatures of 580–600 °C represent at least 90 per cent of the NRM. The residual has unblocking temperatures of 650–680 °C. In other sites (Ni 5, 21, 22, 24, 29, 30, FN 5), both intermediate- and high-temperature components coexist with comparable intensities. Isothermal remanent magnetization curves (IRM) and thermal demagnetization of a three-component IRM generally indicate the presence of both low-coercivity, intermediate-temperature (580 °C) minerals and high-coercivity, high-unblocking-temperature (660–680 °C) minerals in variable proportions (Fig. 3). In the microgranite Ni 33 the low-coercivity, intermediate-temperature minerals account for the major part of the induced magnetization (Ni 33.52, Fig. 3). The same behaviour is observed for the NRM (Ni 33.52, Fig. 2). The carrier of the intermediate-temperature remanence is interpreted as being magnetite, while the high-temperature residual is carried by haematite. Microscopic observations on a polished section of Ni 33 have shown grains of magnetite as well as abundant haematite along cracks and clivage of altered mafic minerals. In site FN 5, the IRM acquisition and demagnetization curves indicate a very small percentage of magnetite against haematite (FN 5.43, Fig. 3). Nevertheless, this small amount of magnetite is responsible for the greatest part of the NRM (FN 5.42, Fig. 2), and haematite is only represented by a very small residual. In site Ni 14, where both magnetite and haematite are evidenced by the IRM procedures (Ni 14.41, Fig. 3), magnetite is also the main carrier of the NRM (Ni 14.41, Fig. 2). On the other hand, in several other sites with IRM curves very similar to that of Ni 14.41, demagnetization of the NRM shows a J/J_0 curve typical of haematite.

Microscopic examination of polished sections showed a predominance of grains of altered minerals consisting of titano-magnetite and ilmenite, more or less oxydized into a haematite and rutile association, and abundant haematite grains along cracks and clivage of amphiboles and biotites (Honnorez, private communication). As a consequence of the presence of haematite, most layered rhyolites, even those with magnetite unblocking temperatures, show a more or less reddish to brownish colour.

Acidic dikes (VM 8 and VN 23), which are also red,

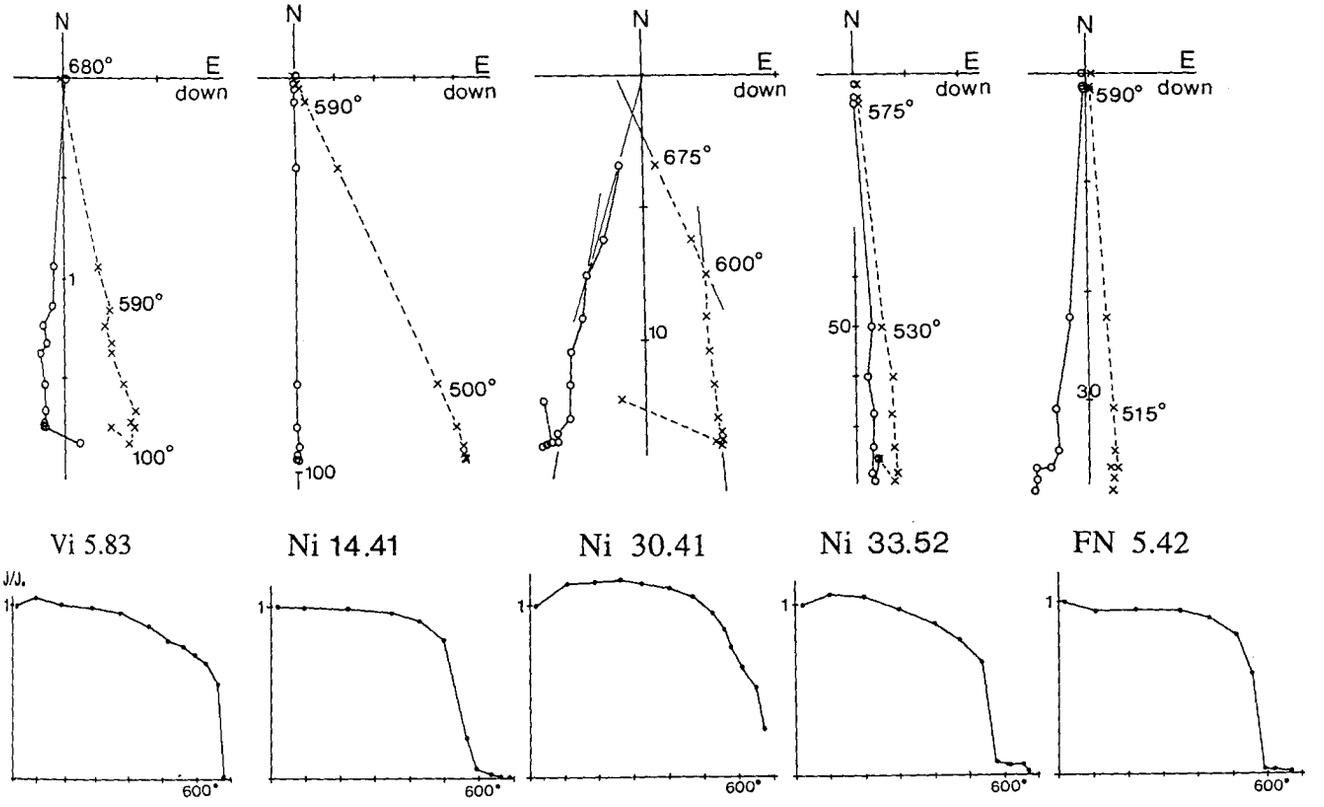


Figure 2. Typical thermal demagnetization of rhyolites. Orthogonal projections and J/J_0 curves. Open dots are in the horizontal plane, crosses in the vertical N–S plane. Intensities are in mA m^{-1}.

display haematite unblocking temperatures. In contrast, the rhyolites VM 9 of the central Vosges and FN 7 of the central Black Forest, which both occur in a granitic environment, have white to violet colours and yield only intermediate unblocking temperatures of magnetite.

Remanence directions

Thermal demagnetization has yielded well-defined characteristic magnetizations with SSW declinations and usually shallow negative inclinations. A low-temperature com-

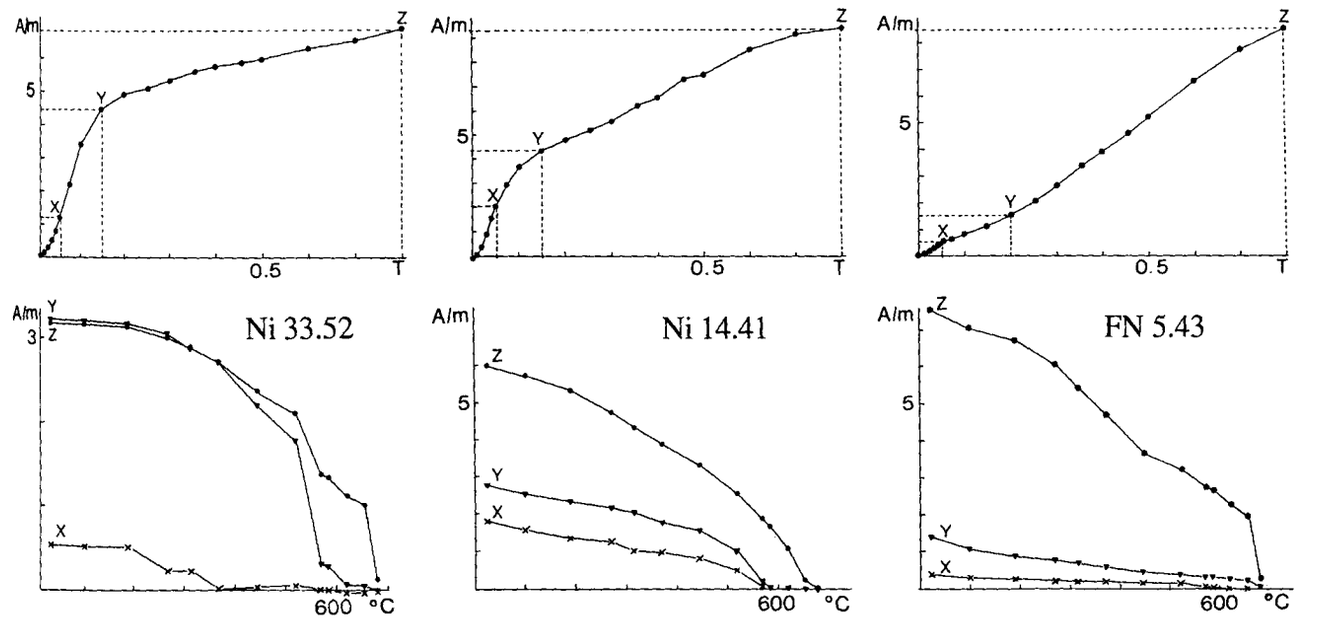


Figure 3. Typical IRM curves and thermal demagnetization of a three-component IRM, after magnetizing the core in a 800 mT field along the Z-axis, 150 or 200 mT along Y and 50 mT along X.

ponent with normal polarity was exhibited in only one site (Ni 5). With the exception of the latter magnetization which may be Triassic, because of its normal polarity, the magnetizations were probably emplaced during the Kiaman reversal, i.e. in the period Late Carboniferous–Late Permian. At the site the characteristic directions usually show nice clusters: the k factor exceeds 50 in all sites. Directions are presented below for the different geographic areas.

The Nideck–Donon volcanic units (Fig. 4, Table 1).

The remanence directions are usually not controlled by lithology and stratigraphy, but depend on the nature of magnetic carriers. According to the IRM curves and thermal demagnetization, the characteristic magnetizations of 28 sites among 33 are carried by haematite. In six sites (Ni 3, 9, 10, 14, 15, 33), at least 90 per cent of the NRM is carried by intermediate-temperature, low-coercivity minerals that are identified as magnetite. When available, the small residuals with higher temperatures usually have the same directions as the magnetite, or directions with slightly higher inclinations (Ni 33, Fig. 4). In seven other sites (Ni 4, 5, 21, 22, 24, 29, 30) two components with comparable intensities and respectively magnetite and haematite unblocking temperatures were separated. The magnetite component generally has a lower inclination than the high-temperature component (Ni 30.41, Fig. 2; Ni 22, 29, Fig. 4). Ni 5, which has nearly the same inclination for both components, is an exception. These three groups have to be examined separately with respect to the tectonic correction.

(1) *Haematite components* (H, Table 1). After tilt correction, the 26 mean directions carried by haematite, for which bedding criteria are available, become significantly more scattered than before correction (Fig. 5). According to McElhinny (1964), the fold test is negative at a 99 per cent confidence level ($k_o/k_c = 2.60 > 2.12$), and the magnetizations are considered as post-tectonic overprints. Because the main tilting occurred prior to the Trias, the mean *in situ* direction NIV3 (187°, −19°, $k = 167$, $\alpha_{95} = 2°$, $N = 26$) has to be corrected (186°, −17°) for the slight bedding (3° towards NW) of the Triassic cover evidenced from contour intervals.

(2) *Intermediate-temperature components* (560–590 °C) (M and MH, Table 1). The fold test is negative at a 95 per cent confidence level and inconclusive at a 99 per cent confidence level ($k_o/k_c = 2.47$, for 13 sites (Fig. 5). M and MH groups both show a different behaviour after tectonic correction.

(2.1) *MH components.* These components coexist with haematite H components, and show a clear negative fold test at a 99 per cent confidence level ($k_o/k_c = 7.83 > 4.16$ for seven sites). Magnetite is secondary as well as haematite and the magnetizations are overprints. After correction for the slight post-Triassic tilting, the mean *in situ* MH direction (184°, −10°, $k = 799$, $\alpha_{95} = 2°$, $N = 7$) becomes (184°, −8°, $k = 799$, $\alpha_{95} = 2°$).

(2.2) *M components.* These components are sometimes associated with a very weak haematite component, and converge after tilt correction (Fig. 5). The McElhinny test (1964) is positive at a 95 per cent confidence level ($k_c/k_o = 4.07 > 2.97$, for six sites), and the magnetizations can be considered as pre-tectonic. Nevertheless, the specimen directions of Ni 3, 10, 14 show comparable, elongated distributions (Fig. 4), which are not compatible with a rapid acquisition of the magnetization during the

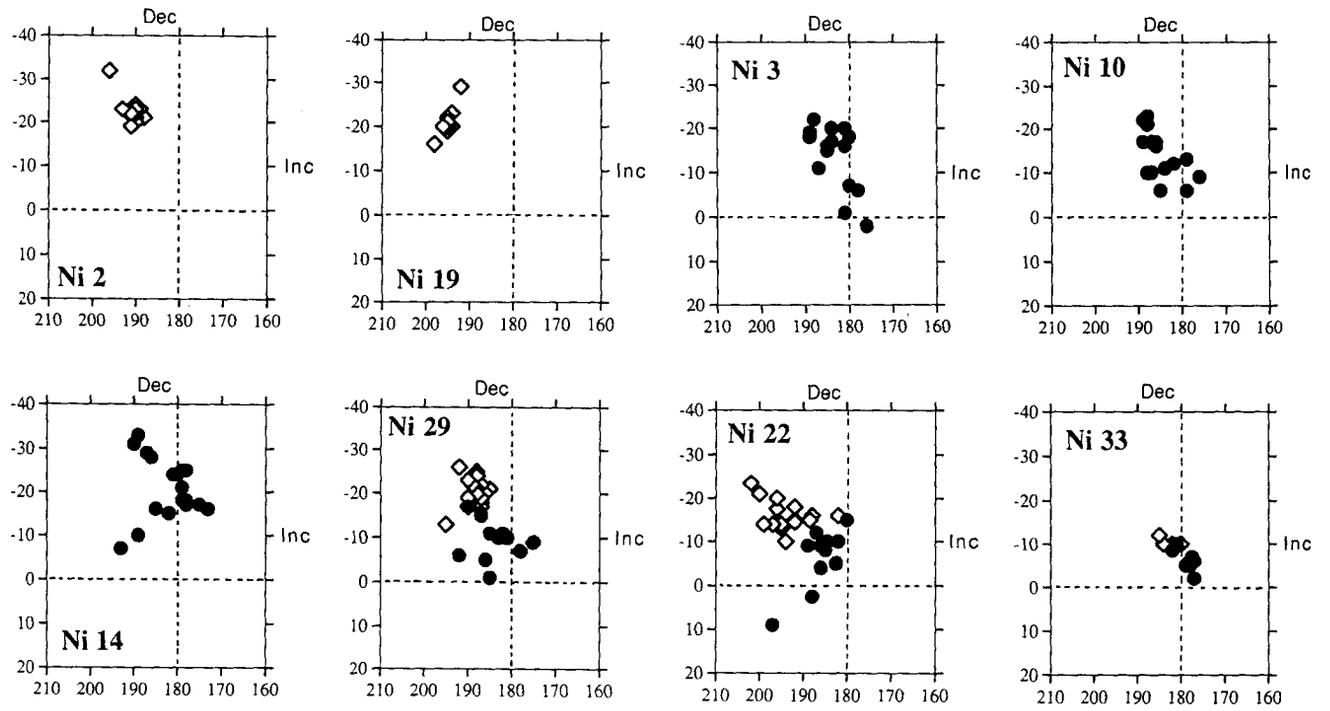


Figure 4. Distribution of the characteristic directions in some typical sites of the Nideck–Donon massif, cylindric projection. ◇ and ● represent magnetizations with high (650-680 °C) and intermediate (580–600 °C) unblocking temperatures, respectively.

Table 1. ChRM in the Nideck–Donon rhyolites (48.50° N, 7.25° E).

Site	Rock type	tb	n	D0	I0	Dc	Ic	k	a95	comments
	Lower unit									
Ni1	rhyol. tuff-breccias	610; 670	6	190	-22	187	-16	2754	1	H
Ni2	ignimbrite	610; 670	6	190	-22	187	-16	1338	2	H
Ni3	rhyol. tuff-breccias	580-610	15	183	-14	182	-10	94	4	M 1+2
		580-610	11	185	-17.5	184	-13	360	2	M 1
		580-610	4	184	-3	184	1	43	4	M 2
Ni4	rhyol. tuff-breccias	580-590;670	6	182	-8	182	4	204	5	MH
		680	3	185.5	-11	185	0	83	14	H
Ni5	rhyol. tuff-breccias	500-530	6	14	30	8.5	28.5	69	6	
		580-600	8	185	-12	183	-9	289	3	MH
		680	12	185	-8	184	-5	138	4	H
Ni6	rhyol. tuff-breccias	680	5	183.5	-20	181.5	-12.5	525	3	H
Ni7	rhyol. tuff-breccias	680	8	182	-22	179	-11	363	3	H
Ni8	rhyol. tuff-breccias	620-680	6	192	-18	191	-4	523	3	H
Ni9	rhyol. tuff-breccias	580-590	14	183	-17	177	-6.5	185	3	M
	Upper unit I									
Ni10	ignimbrite	580-620	15	185	-14	182	-5	143	3	M 1+2
		580-620	7	187.5	-19	183	-10.5	671	2	M 1
		580-620	8	182.5	-10	181	0	273	3	M 2
Ni11	ignimbrite	580-680	7	185	-22	185	-22	336	3	H
Ni12	ignimbrite	580-680	5	182	-15	183	-9	88	8	H
Ni13	lava-like ignimbrite	580; 680	9	192	-18	183	-37	447	2	H
Ni14	lava-like ignimbrite	530-590;680	16	181	-22	178	-6.5	121	3	M 1+2
		530-590;680	4	188	-30	182	-16	886	3	M 1
		530-590;680	12	179	-20	176	-4	279	3	M 2
Ni15	lava-like ignimbrite	580-590;680	8	178	-19	177	-5	250	4	M
Ni16	ignibritic tuff	610-680	6	190	-21	186	-22	58	9	H
Ni17	ignimbrite	680	5	181	-21	180	-12	501	3	H
Ni18	ignimbrite	580-680	8	184	-24	182	-11	172	4	H
Ni19	ignimbrite	610-680	10	195	-21	194	-9	497	2	H
Ni20	ignimbrite	590-630; 680	8	188	-27	184	-21	134	5	H
Ni21	ignimbrite	560-590	4	183	-11	183	4	253	6	MH
		680	7	193	-16	192.5	1	170	5	H
Ni22	ignimbrite	560-580	11	185	-6.5	185	10	121	4	MH
		680	16	194	-15	190	0	153	3	H
Ni23	ignimbrite	680	4	187	-25	184	-22	1605	2	H
Ni24	ignimbrite	580-600	4	181	-10	182	16	868	3	MH
		680	4	185	-17	184	8	442	4	H
	Upper unit II									
Ni25	rhyolitic flow	680	11	184	-15	184	-15	136	4	H
Ni26	rhyolitic flow	680	7	182	-20	180	-9	571	2	H
Ni27	rhyolitic flow	680	4	185	-25	181	-14	651	4	H
Ni28	rhyolitic flow	680	5	176	-19	176	-19	429	4	H
Ni29	rhyolitic flow	560-600	12	184	-10	183	1	149	3.5	MH
		680	16	189.5	-20	186.5	-10	394	2	H
Ni30	rhyolitic flow	560-590	5	187	-12	187	1	224	5	MH
		680	9	190	-19	189	-6	253	3	H
Ni31	rhyolitic flow	680	5	187	-20			1563	2	vertic. foliat.
Ni32	rhyolitic flow	680	7	178	-24			59	8	vertic. foliat.
Ni33	rhyolitic dome	570-590	9	179	-6.5	179	-6.5	661	2	MH
		680	4	183	-11	183	-11	1142	3	H

Table 1. (*Continued.*)

Site	Rock type	tb	n	D0	I0	Dc	Ic	k	a95	comments
	Mean Magnetite M	in situ	6	181.5	-15.5			187	4	
NIV1		corrected	6			179	-7	761	2	45N, 189E
NIV2	Mean Magnetite MH	in situ	7	184	-10	184	-8*	799	2	45N, 182E
		corrected	7			183.5	4	102	6	
	Mean Magnetite M+MH	in situ	13	183	-12.5			233	3	
		corrected	13			181.5	-1	94	4	
	M correct.+ MH corr. for post-Triassic tilt		13		182	-7*		459	2	45N, 184E
NIV3	Mean Hematite H	in situ	26	187	-19	186	-17*	167	2	50N, 178E
		corrected	26			185	-12	64	4	

Mean directions obtained in Permian units. tb: maximum unblocking temperature interval of the considered component and, in some instances, maximum unblocking temperature of the residual with the same direction; n: number of specimens (within a site), or number of sites (within a locality); D_o, I_o: *in situ* direction; D_c, I_c: direction corrected for bedding; comments: label of the considered component and VGP. Means with asterisk correspond to post-tectonic directions, corrected for the post-Permian tilting. H: components carried by haematite. M and MH: components carried by magnetite.

cooling of a common flow. After tilt correction, these distributions overlap (Fig. 6) and fit the results from the microgranitic site Ni 33. Some of the upper part of the distribution is consistent with the mean MH direction, the rest is between the H and MH means. This distribution is not due to incomplete demagnetization because all of the characteristic vectors look perfect on Zijderveld plots (Ni 14.41, Fig. 2). Such an organized scatter of the directions implies that the M magnetizations are not thermo-remanent but are chemical overprints acquired at different stages of an alteration process. The greater scatter in inclination than in declination could be explained by secular variation of the

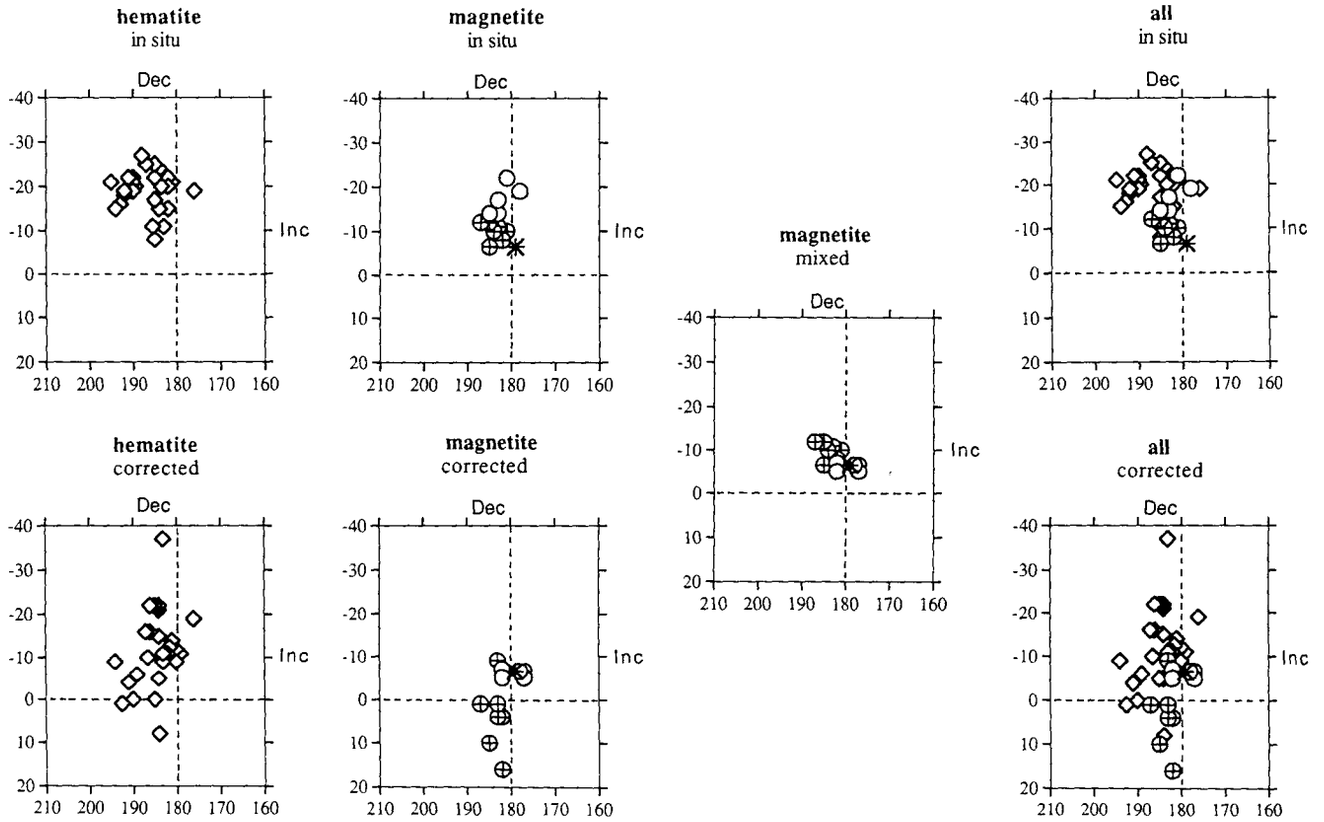


Figure 5. Cylindric projection of the mean directions of high-temperature components H (650–680 °C, ◇) and intermediate-temperature components M and MH (580–600 °C, ○) from Nideck–Donon. ○ represents pre-tectonic M magnetizations; ⊕ shows post-tectonic MH overprints. The star represents the microgranite of Blanc-Rupt.

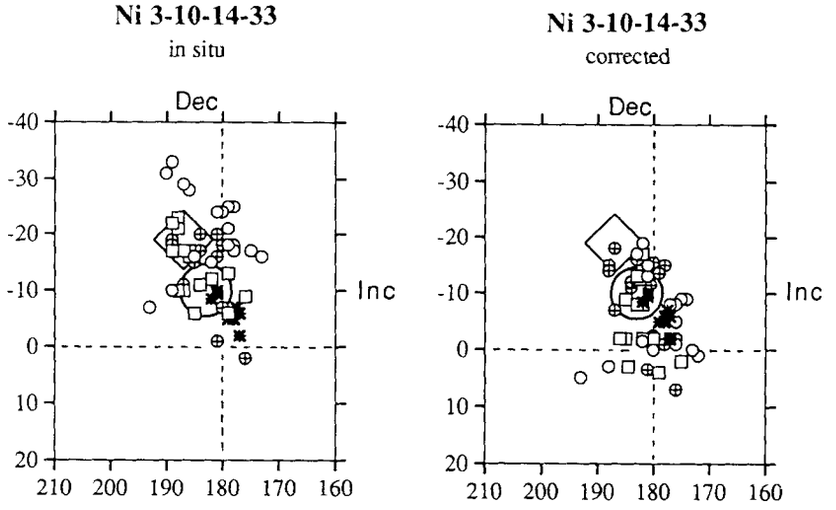


Figure 6. After tilt correction, the elongated distributions of the M directions in sites Ni 14 (○), Ni 10 (□) and Ni 3 (⊕) do overlap. The M directions in the microgranite of Blanc-Rupt Ni 33 (stars) are shown as reference. The large circle represents the mean post-tectonic MH direction and the large diamond shows the mean post-tectonic H direction.

field in the vicinity of the equator during acquisition of the magnetizations. The same trend, however, affects directions from different volcanic layers, and the axis of the distribution is striking up W–down E (Figs 4, 6) instead of up-down. This explanation seems unlikely. Overprinting probably occurred during a geotectonic event, local or global. The positive fold test excludes acquisition of the magnetizations during tilting; thus a latitudinal drift of the Vosges is the most probable solution. As the mean M direction (179°, −7°, $k = 761$, $\alpha_{95} = 2°$, $N = 6$) after tilt correction does not reflect this, the time-dependent evolution was assumed to be a bimodal distribution for sites Ni 3, 10, 14 (Table 1). The two mean directions (180°, −1°, $k = 282$, $\alpha_{95} = 7°$) and (183°, −13°, $k = 770$, $\alpha_{95} = 4°$) give an idea of the latitudinal changes during acquisition of the M components.

On polished sections of rocks with both magnetite and haematite components, irregular grains of titano-magnetite oxidized into haematite and rutile lenses were seen. Secondary haematite grains are abundant in cracks and clivage of altered mafic minerals. As a consequence, the magnetizations in the Nideck–Donon rhyolites result from a more or less continuous oxidizing process of titano-magnetites into, successively, pre-tectonic magnetite, post-tectonic magnetite and finally post-tectonic haematite as well as from crystallization of secondary haematite after the tilting phase.

The Saint Dié basin (Table 2)

Two sites were sampled for AMS investigations. They are located in rhyolites with folded fluidality where bedding criteria are lacking. Nevertheless, because the remanence carrier is solely haematite, the conclusions from the Nideck–Donon rhyolites considering the magnetizations carried by haematite as secondary are probably also valid here. After correction for the slight post-Triassic tilting, the overprints have a mean direction SDV (184°, −20.5°) close to the mean direction NIV3 carried by secondary haematite in the Nideck–Donon units.

The Villé basin (Figs 7, 8, Table 3)

The rhyolitic tuff Vi 3 exhibits two components: a dominating one with unblocking temperatures of magnetite, characterized by shallow inclinations (195.5°, −0.5°, $k = 96$, $\alpha_{95} = 5°$), and a high-temperature residual with significantly higher inclinations (193°, −23°) (Figs 7 and 8). A thin red-brown volcanic-sedimentary layer separates the rhyolitic tuff Vi 3 from the ash-tuffs Vi 1–2, 4 which occupy the lower position. The white-grey ash-tuffs are partly mottled red and become almost red in the upper part of the layer. The pigmentation is considered as allogenic and comes from the dark, reddish, overlying volcano-sedimentary layer. The white-grey specimens are very weakly magnetized and contain no characteristic magnetizations. The intensity of the NRM increases with the reddish pigmentation. Consequently, the high-temperature characteristic magnetizations are carried by the secondary haematite responsible for the pigmentation. The directions of characteristic magnetizations are much more scattered than in other volcanic rocks (Fig. 8). They seem to migrate from the primary direction observed in the overlying rhyolitic tuff Vi 3 (195.5°, −0.5°) towards a more easterly direction (170°, −10°), consistent with the intermediate-temperature

Table 2. ChRM in rhyolites from the Saint dié Basin (47.32° N, 6.85° E).

Site	Rock type	tb	n	D0	I0	Dc	Ic	k	a95	comments
SD2	rhyolitic flow	590-680	5	182	-21			113	7	H bedding ?
SD3	rhyolitic flow	590-680	7	188	-30			283	4	H bedding ?
SDV	Mean Hematite		2	185	-26	184	-20.5*			53N, 180E

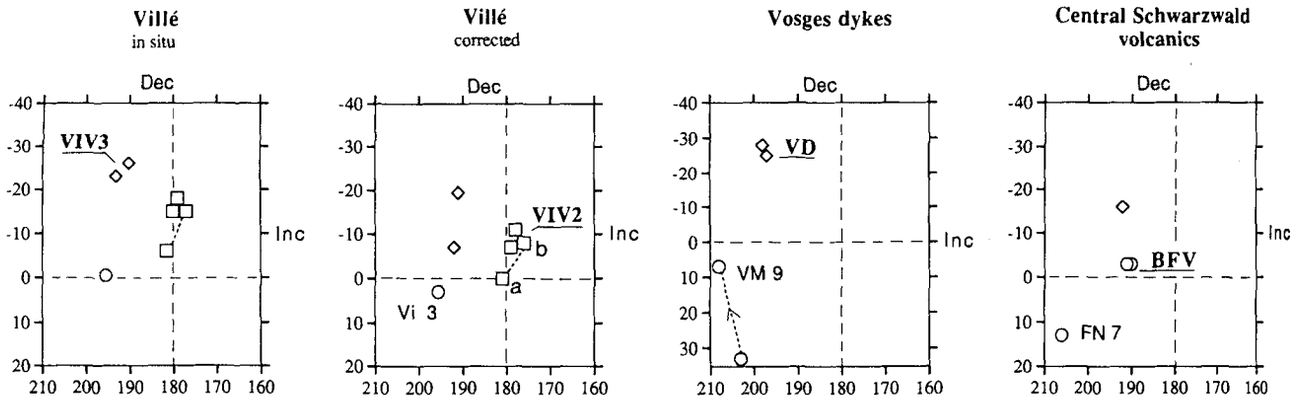


Figure 7. Mean directions of sites from the Villé basin and surroundings and from the Black Forest. For site Vi 1–2 two mean directions are represented, connected with a dashed line: 'a' in slightly pigmented tuffs and 'b' in full reddish tuffs. The pre- and post-tectonic directions of rhyolite VM9 are connected with a small circle. Symbols are the same as for Fig. 4. Tables 3–5 give the means corresponding to the labelled groups of directions.

directions of Nideck–Donon, and then towards the secondary direction of Vi 3 (193°, −23°). This latter direction is very close to the *in situ* direction of the post-volcanic micro-conglomerate Vi 6 in which the magnetization is also secondary, because the same direction was found in fine-grained specimens as in specimens with centimetric clasts. Both magnetizations are considered as post-tectonic overprints. A mean direction was calculated with the primary magnetization of Vi 3 and the overprint of Vi 1–2 corresponding to those of the less-impregnated ash-tuffs. After tilt correction the mean is (188°, 1.5°, $k = 60$, $N = 2$), which is comparable with the mean computed with the shallow M directions of Ni 3, 10, 14 (180°, −1°, $k = 282$, $\alpha_{95} = 7°$). Concerning the characteristic magnetizations in ash-tuffs, which are secondary, it is unclear whether or not they have to be corrected for the slight tilting of 8° towards the NW ($k_c/k_o = 992/1271$). A weak preference is given to the pre-tectonic direction (178°, −9, $k = 992$, $\alpha_{95} = 4°$, $N = 3$) because of its consistency with the NIV1 mean of the Nideck–Donon volcanics (179°, −7°, $k = 761$, $\alpha_{95} = 2°$, $N = 6$) and the microgranite of Blanc-Rupt (179°, −6.5°, $k = 661$, $\alpha_{95} = 2°$, $n = 9$).

Intrusives around the Villé basin (Fig. 7, Table 4)

The acidic dikes VN 23 (Edel *et al.* 1986) and VM 8 show a reddish pigmentation due to haematite, as evidenced by the high unblocking temperature of the magnetizations. The very close directions (VD mean: 197.5°, −26.5°, $k = 1340$) are similar to those of the late overprints of the

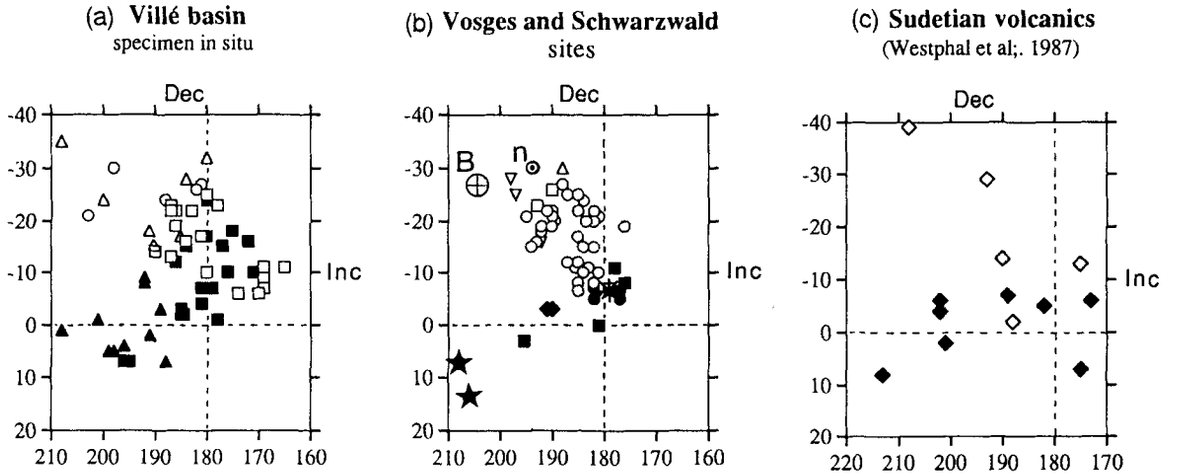


Figure 8. (a) Characteristic *in situ* directions in specimens of rhyolitic tuffs (triangles), ash-tuffs (squares), and conglomerate (circles) from the Villé basin. Open symbols represent late overprints in red ash-tuffs, rhyolitic tuffs and conglomerates. Full symbols indicate primary directions in rhyolitic tuffs and early overprints in white and mottled ash-tuffs; (b) mean site directions in the Vosges and Black Forest. Open symbols represent post-tectonic overprints carried by haematite and magnetite (circles: Nideck–Donon rhyolites). Full symbols indicate pre-tectonic magnetite components (stars: Late Carboniferous rhyolites; squares: rhyolites and tuffs; large star: microgranite; diamonds: Black Forest rhyolites). 'n' is the sole reversed direction with normal polarity (Ni 5), 'B' is an Anisian (Early Triassic) direction from northern Vosges sandstones (Edel 1993). (c) Pre-tectonic (full symbols) and post-tectonic directions (open symbols) in Late carboniferous–Early Permian volcanics of the Polish Sudetians (Westphal *et al.* 1987).

Table 3. ChRM in volcanic units from the Villé Basin (the central Vosges) (48.33° N, 7.33° E).

Site	Rock type	tb	n	D0	I0	Dc	Ic	k	a95	comments
Vi1-2	white+red mottled tuffs	680	9	181.5	-6	181	0	83	6	H1
	red ash tuffs	680	9	177	-15	176	-8	52	7	H2
	all ash tuffs	680	18	179	-11	178	-4	56	4	
Vi 4	red+ red mottled ash tuffs	680	6	179	-18	178	-11	154	5	H2
Vi 3	rhyolitic tuffs	580-610	11	195.5	-0.5	195.5	3	96	5	MH 38N,167E
		630-680	8	193	-23	191	-19.5	39	9	H3
Vi5	white+ red mottled tuffs	580	1	173	-7	173	2			
		660-680	9	180	-15	179	-7	140	4	H2
Vi6	microconglomerate	680	5	190	-26	192	-7	74	9	H3
VIV1	Mean MH+H1	580-680	2	188.5	-3			58	33	
	(early magnetizations)					188	1.5	60	33	40N, 177E
VIV2	Mean H2	660-680	3	179	-16			1271	3	
	(hematite impregnation)					178	-9	992	4	46N, 190E
VIV3	Mean (late overprint)	680	2	191.5	-25			798	9	53N, 168E
						191.5	-13	84	28	

Table 4. ChRM in dikes around the Villé basin (48.25° N, 7.25° E).

Site	Rock type	tb	n	D0	I0	Dc	Ic	k	a95	comments
VN23	acidic dyke	675	4	198	-28			341	5	Edel et al., 1986
VM8	acidic dyke	>650	5	197	-25			63	10	
VM9	rhyolite (Stephanian ?)	580	10	203	33	208 ?	7 ?	64	6	33N, 153E
VD	Mean (high temperature)		2	197.5	-26.5			1340	7	53N, 158E

neighbouring Villé basin (VIV3: 191.5°, −25°, $k = 798$). The result from the white to violet rhyolite outcropping within the Crêtes granite (VM 9, Fig. 1, 4, 5) is very different, showing a positive inclination. The intermediate unblocking temperature of 580 °C indicates magnetite.

Volcanics of the central Black Forest (Fig. 7, Table 5)

The three sites FN 5, 6 and 7 were investigated in a previous study (Edel 1987). They are presented here after additional demagnetizations and IRM processing. Two flat-lying rhyolitic flows (FN 5, 6), about 10 km apart, yielded the same direction (191°, −3°) for the intermediate-temperature (590 °C) component (Fig. 7). A high-temperature residual component with higher inclinations (192°, −16°) is also evidenced in site FN 5. The rhyolite FN 7 has the same characteristics as the rhyolite VM 9 of the central Vosges: a white to violet colour and magnetite unblocking temperatures. The direction with positive inclination is also similar (206°, 13°).

DISCUSSION OF THE RESULTS

Previous investigations in the northern Vosges

Investigations on the Nideck–Donon volcanics by Roche *et al.* (1962) yielded a mean direction (192°, −13°) not very different from the mean (184°, −13°) calculated with the mean intermediate-temperature direction NIV1 (179°, −7°) and high-temperature direction NIV3 (186°, −18°) from this study. The deviation is due to the fact that some of the samples were not sufficiently demagnetized and to the lack of tilt correction. Some years later, Westphal (1972) showed a shift from shallow to higher inclinations that he interpreted as reflecting significantly different ages of emplacement of the lower and upper units (Late Carboniferous–Early Permian and Later Permian, according to Mihara 1935). The lack of tectonic correction and thermal demagnetization precludes inclusion of those results in this study. Nevertheless, the present work confirms the shift of

Table 5. ChRM in rhyolites of the central Black Forest (48.5° N, 8.0° E).

Site	Rock type	tb	n	D0	I0	Dc	Ic	k	a95	comments
FN5	rhyolitic tuff	580	8	191	-3	191	-3	252	3	M
		>600	6	192	-16	192	-16	192	5	H 49N, 170E
FN6	rhyolitic tuff	580-670	10	190.5	-3	190.5	-3	287	3	M
FN7	rhyolite (Stephanian?)	580	7	206	13			94	6	31N, 157E
BFV	Mean Magnetite M		2	191	-3	191	-3	52429	1	M 42N, 173E

the inclinations and demonstrates that it is the magnetizations that were acquired at different ages. K–Ar dating of biotite yielded similar ages for the lower (297 Ma, Schneider *et al.* 1994) and the upper (299 Ma, Lippolt & Hess 1983) units. This age is the lower limit for acquisition of the shallow inclination components carried by magnetite.

Permian overprinting

The new results show that the majority of the Early Permian volcanic rocks from the northern Vosges are overprinted. A first group of directions (NIV3, VIV3, SDV, VD; mean VGP: 52.5° N, 171° E, $k = 170$, $\alpha_{95} = 7°$) is characterized by: (1) high inclinations (−10° to −26°); (2) a negative fold test in the Nideck–Donon units and the Villé basin; and (3) haematite as carrier of the remanence. IRM and NRM demagnetization curves, and microscopic analysis of polished sections indicate the predominance of secondary haematite and haematized titano-magnetites. The association of haematite, ilmenite and rutile grains seen in thin sections indicates hydrothermal alteration. The results of the microscopic analysis are very similar to those made for the coeval rhyolites of the Saar-Nahe basin in western Germany (Berthold *et al.* 1975). The Permian hydrothermal alteration can be considered to be a wide-spread feature.

The second group consists of three very similar directions from Nideck–Donon and the Villé basin (NIV1, NIV2 and VIV2; mean VGP: 45° N, 187° E) with shallower inclinations (mean $I = -8°$) than for the group characterized by haematite (mean $I = -18°$). In the Nideck–Donon formations, the remanence carrier has the characteristics of magnetite. The negative fold test of the MH group (NIV2) indicates that magnetite is clearly secondary when strong haematite components also occur. Despite the positive fold test of the M group, the elongated distribution of the directions also indicates secondary acquisition of the magnetizations during an apparent shift of the magnetic field. Magnetite also results from an alteration process, probably oxidation of titano-magnetite. The tilting phase occurred during this alteration, while haematite crystallized afterwards. The VIV2 component from the Villé basin is clearly an overprint, resulting from impregnation of the ash-tuffs by allogenic iron oxides originating from the overlying volcanic-sedimentary layer.

The intermediate-temperature direction found in Vi3 and the early overprints in white and mottled tuffs of Vi 1–2 (Mean VGP: 40° N, 177° E) are older. Both are consistent with the intermediate-temperature magnetizations of sites FN 5 and FN 6 in the Black Forest (VGP; 42° N, 173° E), which are considered as primary and which are dated at 287 Ma (Lippolt *et al.* 1983). According to the positive inclinations and the Rb–Sr age of 297 Ma obtained on FN 7 (Lippolt *et al.* 1983), the magnetizations carried by magnetite in rhyolites VM 9 and FN 7 which both extruded Visean granites, are the oldest from the Late Variscan magmatic rocks (VGP: 32° N, 155° E). With Vi3, they may be the sole primary magnetizations of the northern Vosges.

The trend of the remanence directions

The distribution of all mean directions obtained in the central and northern Vosges and the central Black Forest and the distribution of the directions of the specimens of tuffs from the Villé basin look very similar (Fig. 8). The general trend indicates an increase in the inclinations from shallow positive and negative for the oldest directions to negative, up to −25°, for the post-tectonic overprints. The few data with shallow inclinations show a W–E trend of the declinations, which suggests a clockwise rotation of the units (directions in black, Fig. 8). After a cusp at about (180°, −8°) which coincides with the tilting phase that separated the acquisitions of M and MH overprints, the directions move up and westwards, i.e. during overprinting of the MH and H components, the Vosges and consequently western Europe were drifting towards the north and rotating counterclockwise. A similar distribution was observed in probably contemporaneous rhyolitic units of the Sudetians, where the W–E trend of the pre-tectonic directions is followed by a W–E and upward trend of the overprints (Fig. 8) (Westphal *et al.* 1987).

THE POLAR WANDER PATH OF VARISCAN EUROPE

The available data for Variscan Europe

The bend evidenced in the distribution of the palaeomagnetic directions (Fig. 8) appears to be a general feature when looking at the VGP map obtained with published data from Variscan Europe (Fig. 9). This is not obvious on published APWP maps (Irving & Irving 1982; Van der Voo 1990) because of successive smoothing due to (1) calculations of mean directions within a given area by various researchers, and (2) calculations of mean poles for a given time-span. Another reason is that a large part of the Permian data is rather old, and the mean poles from Nideck–Donon (Roche *et al.* 1962), the Black Forest (Konrad & Nairn 1972), Saar-Hahe (Nijenhuis 1961), Thuringian Forest (Mauritsch & Rother 1983), the western Bohemian Massif (Soffel & Harzer 1991), and the central and eastern Bohemian Massif (Birkenmaier *et al.* 1968) probably represent mostly secondary magnetizations acquired at different stages of the Permian, as in the northern Vosges. When possible, the data should be re-examined, separated into primary and secondary magnetizations, and corrected for tilt when necessary. For instance, the pole from the western Bohemian Massif (42° N, 166° E; Soffel & Harzer 1991) is the mean of two groups of directions: (1) directions in layered porphyries from the Franconian Line characterized by intermediate and high unblocking temperatures and shallow inclinations after tectonic correction (recalculated VGP: 38° N, 170° E), and (2) *in situ* directions in intrusives from the western Bohemian Forest characterized by high unblocking temperatures and significantly higher inclinations (−21°) (recalculated VGP: 47° N, 159° E). The first pole is consistent with Autunian poles, the second falls into the group of Thuringian poles. In southern France, the scatter of the poles obtained by various teams in Autunian sediments from the Lodève basin is partly explained by an inconclusive fold test, assumed to be also related to problems of relative timing of magnetization acquisition and bed-tilting processes (Cogné *et al.*, 1990).

Despite the lack of optimal data with a positive fold test, the Autunian poles are mainly characterized by latitudes around 40° corresponding to shallow positive and negative

inclinations of the magnetizations. The Saxonian–Thuringian poles display higher latitudes, around 50°, and higher negative inclinations. As a consequence, in the Lodève basin, where all investigated sites are dipping towards the south, confidence is given to directions with positive inclination after dip correction (Kruseman 1962; Merabet & Guillaume 1988). Those with negative inclinations are considered to be syn/post-tectonic overprints. The large scatter of the longitudes/declinations is due to local rotations during the Permian (see later).

Ages of the magnetizations

The radiometric dates obtained on the Nideck–Donon rhyolites represent cooling ages of biotites i.e. ages of emplacement of the volcanics. As the rhyolites were completely remagnetized after cooling, an attempt has been made to date the magnetizations by correlations with published dated poles that are considered to be reliable (Fig. 9).

Using this method, the oldest magnetizations in rhyolites from the Vosges and Black Forest are characterized by positive inclinations with magnetite as carrier. The absence of haematite supports the argument that the magnetizations are primary. The mean pole VSR corresponding to the rhyolites VM 9 and FN 7 falls into the Westphalian–Stephanian group of poles (Fig. 9). Relevant rock ages (FLG, TPG, MMI, BFV, Table 6) range from 296 ± 5 Ma, which is a Rb–Sr date for FN 7 (Lippolt *et al.* 1983), to 308 ± 7 Ma found for the Flamanville granite with the same method (Vidal 1980). A mean age of 303 ± 12 Ma corresponding to the Stephanian (Hess & Lippolt 1986) is assigned to the mean VGP labelled 'WS' and computed from Table 6.

Poles relative to magnetizations considered as 'primary' in the Villé basin (Vi 3), Black Forest (BFV) and Pfalz-Odenwald-Spessart (PSOV, Table 6) are in good agreement with results obtained on Autunian sediments from the Lodève and Saint-Affrique basins (Kruseman 1962; Merabet & Guillaume 1988; Diego-Orozco & Henry 1992), the Bohemian Massif (Krs 1968) and from Late Carboniferous–Early Permian volcanites from Saar-Nahe (Berthold *et al.* 1975), the Bohemian Massif (Mauritsch & Rother 1983; Soffel & Harzer 1991) and the Sudetians (Westphal *et al.* 1987). The tuffs of the Villé basin are considered to be equivalent to the tuffs and breccias from the lower Nideck–Donon unit which are dated at 299 ± 9 Ma by K–Ar processing and 294 ± 1 Ma by the Ar–Ar plateau on biotite (Lippolt & Hess 1986; Montigny, in Schneider *et al.* 1994). These convergent ages are interpreted as crystallization ages due to the rapid cooling of the rhyolites. When it is assumed that the intermediate-temperature component of Vi 3 (VGP: 38° N, 167° E) is thermoremanent, the radiometric ages may correspond to

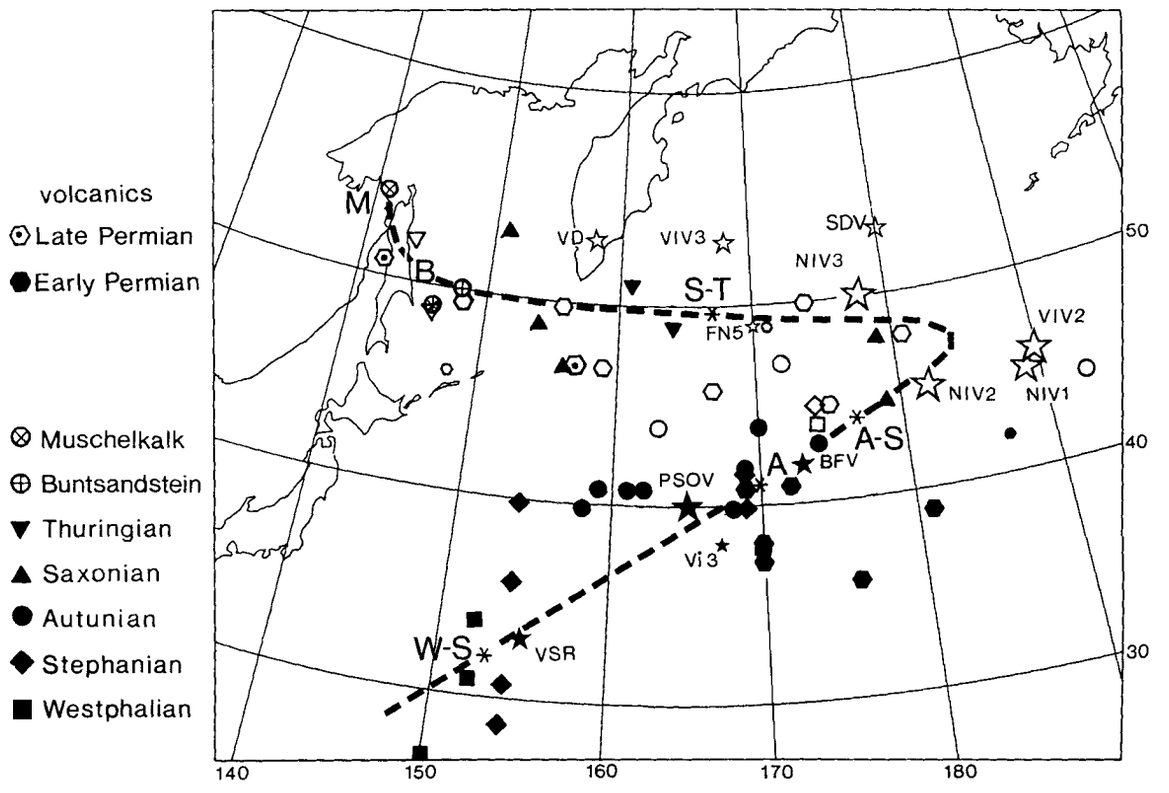


Figure 9. Virtual Geomagnetic Poles from Variscan Europe (Table 6). Stars represent results from this study (Tables 1–5). Open symbols are Saxonian–Thuringian overprints. Small, intermediate, large symbols represent respectively 1, 2 and $N > 2$ sites. The APWP (thick dashed line) is based on the mean poles W–S (Westphalian–Stephanian), A (Autunian), A–S (Autunian–Saxonian), S–T (Saxonian–Thuringian), represented by stars (Table 7) and computed from Table 6, and on the poles B (Buntsandstein) and M (Muschelkalk) (Table 6).

Table 6. Late Carboniferous to the Late Triassic virtual geomagnetic poles of Variscan Europe.

symb	locality	rock type	rock age	magnetization age	age Ma	lat °N	long °E	n or N	k or K	α95 A95	is/c	Mean	authors
	SW England												
EV	Exeter	traps	Autun., 279 Ma	L. Thuringian?	245?	49	148	4	211	4	c		Zijderveld, 1967
	S Norway												
OV	Oslo graben	tholeiit. alkali-basalts	E. Permian	L. Permian?		47	157		21	1			Van Everdingen, 1960
	W. France												
FLG	Flamanville	granite	Westph. 308 Ma	Westph., 308 Ma	308	27	150	8	95	6	sc	WS	Cogné,1988
TPG	Tregastel-Ploum.	granite	Westph. 306 Ma	Westph., 306 Ma	306	34	152	5	121	7	is	WS	Duff, 1979
	S. France												
LOAa	Lodève	red + grey sandst.	Autunian	Autunian	285	44	170		517	2	c	A	Kruseman, 1962
LOSa	"	red Sandstones	Saxonian	Saxonian	265	47	156		33	5	c	ST	"
LOAb	"	ciner. red sandst.	Autunian	Autun.-Saxon?	270	49	171	2	291	15	c	AS	Courtillot et al. 1986
LOAc	"	red + grey sandst.	Autunian	Autunian	285	42	169	8	517	2	c	A	Merabet, Guillaume, 1988
LOSb	"	red sandstones	Saxonian	Saxonian	265	49	154	6	2096	1	c	ST	"
LOSc	"	red sandstones	Saxonian	Saxon.-Thur.	258	53	151	1	122		c	ST	Evans and Maillol, 1986
LOAd	"	red + grey sandst.	Autunian	Autun.-Saxon.	270	44	163	3	282	7	is+c	AS	Cogné et al., 1990 *
SAAa	St. Affrique	grey sandst.	Autunian	Autun.-Saxon.	270	44	193	3	101	12	is	AS	Cogné et al., 1990 *
SAAb	"	pellites, sandst.	L. Autunian	L. Autunian	275	41	161	6	93	6	c	A	Diego-Orozco, Henry, 1992
SASa	"	sandstones	E. Saxonian	E. Saxonian	270	45	179	7	393	3	c	ST	"
SASb	"	sandstones	Sax.- Thur.	Saxon.- Thur.	258	48	179	8	190	4	c	ST	"
SAT	"	sandstones	Thuringian	Thuringian	250	49	164	8	114	5	c	ST	"
	SE France												
MAV	Maures	siltstones	Thuringian	Thuringian	250	51	161	10	160	4	c	ST	Merabet, Daly, 1986
ESS	Esterel	sediments	Saxon.-Thur.	Saxon.-Thur.?	264?	49	146	3	356	7	c	ST	Zijderveld, 1975
ESV	Esterel	rhyolites, basalts	Saxon. 278-264	Saxon.-Thur.?	264?	51	142	7	112	6	c	ST	"
DBS	Dôme de Barrot	red beds	E. Thuringian	E. Thuringian?	255?	52	144	46	66	3	c		Van den Ende, 1977
	Central France												
MMI	Montmarault	intrusives+hostrocks	L. Carboniferous	L. Carboniferous	300	29	154	5	25	16	is	WS	Edel, 1987
	NE France												
MOV1	Moissey, Jura	ignimbr. tuffs	Autun.- Saxon.	Autun.- Saxon.	270	41	172	7	123	5	c	AS	Thompson et al., 1986
NIV	Nideck-Donon	*ignimb, tuffs, flows*	*Autun., 298*	*Permian*	270?	47	169	9			is		Roche et al., 1962 *
NIVa	Nideck	ignimb. tuffs	Autun., 298	Autunian	298	36	162	1			is		Westphal, 1972 *
NIVb	"	"	"	"	298	41	164	1			is		"
NIVc	"	"	"	Saxon.- Thur.	258	48	180	1			is		"
NIV12	Nideck-Donon	ignimb, tuffs, flows	Autun., 298	Autun.- Saxon.	270	45	184	13	459	2	is+c	AS	this work
NIV3	"	"	"	Saxon.-Thur.	258	50	178	26	167	2	is	ST	"
SDV	St. Dié basin	rhylitic flows	Autun., 298	Saxon.-Thur.	258	53	180	2	119		is,c	ST	"
VIV1	Villé basin	tuffs	Autunian	Autunian	298	40	177	2	60		c	A	"
VIV2	"	ash tuffs	Autunian	Autun.- Saxon.	270	46	190	3	992	4	c	AS	"
VIV3	"	tuffs, conglomerate	Autun.-Saxon.	Saxon.-Thur.	258	53	168	2	798		is	ST	"
VD	Central Vosges	acidic dykes	L. Carb.- Perm.	Thuringian ?	250?	53	158	2	1340		is	ST	"
VM9	Central Vosges	rhyolite	L. Carb.-Perm.	Stephanian?	300?	33	153	1			is	WS	"
NVS	Northern Vosges	sandstones	Scyth.-Anisian	E. Anisian	240	50	148	4	614	3	0	B	Edel, 1993
HES	Heming	limestones	Anisian- Ladin.	Anisian- Ladin.	235	54	141	1	97	3	O	M	Théveniaut et al., 1992

Table 6. (*Continued.*)

symb	locality	rock type	rock age	magnetization age	age Ma	lat °N	long °E	n or N	k or K	α95 A95	is/c	Mean	authors
	SW Germany												
FN7	Centr. Schwarzw.	rhyolite	St.-Aut. 296 Ma	Steph.-Autun.	296	31	158	1			is	R	Edel, 1987
VSR	Vosg.- Schwarzw	rhyolites	Stephanian?	Stephanian?	300?	32	155	2	847		is	WS	FN7+VM9
BFS	Schopfheim bas.	red shale	Oberrotliegende	Stephanian?	300?	31	154	5	76	9	?	WS	Konrad, Nairn, 1972
BFP	S. Schwarzwald	quartz porphyr. dykes	Steph. - Autun.	Steph.- Autun.	295	42	187	2	66		?	A	"
BFVa	Schwarzwald	porphyr., ignimbr. flows, sedim.	Steph.- Autun.	Autun.- Saxon.?	270?	50	174	12	26	9	?		"
BFVb	Centr. Schwarzw.	rhyol. ignimbrites	Autun., 287 Ma	Autunian	287	42	173	2	52429		0	A	this work
PFVa	Pfälzer Wald	melaphyr., sandst.	Autun.- Saxon.	Autun.- Saxon.	275?	45	175	4	187	7	?		Konrad, Nairn, 1972
PFVb	Pfälzer Wald	basalt	Autun.- Saxon.	Autun.- Saxon.	275	39	160	1			0		Edel, Wickert, 1991, revis.
PFVc	"	"	Autun.- Saxon.	Saxon.-Thur.	258?	46	148	2			0		"
SPV	Spessart	ignimbrite	Steph. - Autun	Steph. - Autun.	295	41	164	1			0		"
ODV	Odenwald	ignimbrite	Autunian	Autunian	291	42	173	1			is		Konrad, Nairn, 1972
PSOV	Pfalz, Odenwald, Spessart	basalt, ignimbrite	Autunian	Autunian	295	40	165	3	237	8	0	A	PFVb+SPV+ODV
NAV	Nahe	rhyol., intrus.	Autun., 297 Ma	Autunian	297	46	167		54	13		A	Nijenhuis, 1961
SNV	Saar-Nahe	rhyol., intrusions	Autun., 297 Ma	Autunian	297	41	169	11	9	16		A	Berthold et al., 1975
	E. Germany												
TWS	Thüringerwald	red beds	Autun. - Saxon.	Autun.- Saxon.?	270	46	163	10	10	6	0		Mauritsch, Rother, 1983
			"	Saxon.-Thur.	258	47	172	4	56	12	0	ST	", extract
TWV	Thüringerwald	porphyries	Autun.- Saxon.	Autun.- Saxon.	270	37	170	9	10	7	is	AS	Mauritsch, Rother, 1983
WBM	western margin	porphyr., granit., silt	L. Carb.- E. Perm	L. Carb.- E. Perm	295?	42	166	9	65	6	c		Soffel, Harzer, 1991
WBM1	Bohemian Massif	"	"	L. Carb.- E. Perm	295	38	170	5	40	12	c	A	", extract
WBM2	"	"	"	Sax.- Thur.?	250?	47	159	4	201	6	c	ST	", extract
BMS	E. Germany	sand/limestones	Bunt.-Muschelk.	Bunt.-Muschelk.	240	49	146		14	15		B	Rother, 1971
	Bohemia												
KRS	Kladno-Rakovnik	red mudstones	Westph.- Steph.	Westph.-Steph.	305	36	154	4	42	3	0	WS	Birkenmajer et al., 1968
PBS	Plzen basin	red sandstones	Westph.- Steph.	Westph.- Steph.	305	31	152	3	13	5	0	WS	"
ISWa	Inner Sudetic B.	sandstones, volc.	Westph.- Steph.	Steph.-Autun.?	295?	36	176	10	12	5	0	A	"
BLSn	Blanice graben N	red sediments	Stephanian	Steph.-Autun.	295	40	154	25	91	3	0	A	Krs, 1968
BOSs	Boskovice grab.	red sediments	Stephanian	Steph.-Autun.	295	40	169	40	34	4	0	A	"
SKS	Sub-Krkonose B.	red sediments	Stephanian	Steph.-Autun.	295	45	174	57	31	3	0	A	"
BLAn	Blanice graben N	red sediments	Autunian	Autunian	285	40	158	43	38	4	0	A	"
BLAs	Blanice graben S	red sediments	Autunian	Autunian	285	41	162	34	154	2	0	A	"
BOAs	Boskovice gr. S	red sediments	Autunian	Autunian	285	40	168	40	53	3	0	A	"
BOAc	Boskovice gr. C	red sediments	Autunian	Autunian	285	41	159	45	37	4	0	A	"
SKAS	Sub-Krkonose B.	red sediments	Autun-Saxon.	Autun.-Saxon.	275	42	169	116	14	4	0	A	"
SKA	Sub-Krkonose B.	red sediments	Autunian	Autunian	285	38	170	89	30	3	0	A	"
	SW Poland												
ISWb	Inner Sudetic B.	porphyritic effus.	Westph.- Steph.	Steph.-Autun.?	295?	44	174	10	18	4	?		Birkenmajer et al., 1968
ISV1	In. Sudetic Bas.	porphyritic effus.	Westph.- Steph.	Steph.-Autun.	295	39	181	8	27	11	c	A	Westphal et al., 1987
ISV2	In. Sudetic Bas.	porphyritic effus.	Westph.- Steph.	Saxon.-Thur.	258	48	181	5	18	18	is	ST	"

VGP published for Variscan Europe including this study. Lab refers to the age interval assigned to the VGP taken into account in the calculation of mean VGP (Table 7). is/c indicates respectively *in situ* and tilt-corrected directions, 0 represents flat-lying units. Asterisk indicates recalculated poles.

acquisition of the magnetization. Rb–Sr ages of 287 ± 5 Ma were obtained for the rhyolites FN 5 and FN 6 in the Black Forest (mean VGP BFVb: 42° N, 173° E) (Lippolt *et al.* 1983). As the mean Autunian pole A (Fig. 6; Table 6) lies between Vi 3 and BFVb, an intermediate age of 292 ± 13 Ma is assigned to VGP 'A'.

The poles VIV2 from the Villé Basin and NIV1–NIV2 from Nideck–Donon represent the first overprints and mark the bend in the distribution of the palaeomagnetic directions (Fig. 8). They are located at the extremity of the APWP hair-pin (Fig. 9). The closest pole to NIV1 and NIV2 is an Early-Saxonian pole from the Saint-Affrique basin (Diego-Orozco & Henry 1992). The inclinations/latitudes of NIV1, NIV2 and VIV2, between those of the Autunian and Saxonian–Thuringian, are also in favour of a Late Autunian–Early Saxonian age of 272 ± 10 Ma for acquisition of the magnetizations. Another argument in favour of this age is the tilting phase, which occurred after acquisition of the M magnetizations (NIV1) and prior to acquisition of the MH magnetizations (NIV2). This tectonic phase is not dated in the northern Vosges. But in the Saar-Lorraine basin, 60 km farther north, the principal tectonic feature is the discordance of the Saxonian on the Autunian (Donsimoni 1981).

The late overprints carried by secondary haematite from the Villé basin (VIV3), Nideck–Donon (NIV2), Saint-Dié basin (SDV) and Vosges dikes (VD) (mean VGP: 52.5° N, 171° E) agree with the Saxonian–Thuringian poles of southern France (Kruseman 1962; Merabet & Guillaume 1988; Merabet & Daly 1986; Diego-Orozco & Henry 1992). Consequently, the hydrothermal alteration responsible for the haematization of titano-magnetites and crystallization of secondary haematite lasted from Early Saxonian to Late Thuringian and probably up to the Early Triassic, as suggested by the existence of a secondary magnetization with normality polarity in Ni 5. A mean age of 258 ± 15 Ma is assumed for the mean 'SA' pole (Table 7).

Interpretation of the mid-Permian hair-pin

We argue that the sharp bend in the pole distribution is due to the superposition of two tectonic features which are probably related.

(1) *Local and regional rotations.* The Autunian poles show an elongated E–W distribution due to a scatter in declinations, while the inclinations remain rather stable and shallow (Bohemian Massif and Saint-Affrique basin). A comparable distribution can be seen for the Saxonian–Thuringian data, but with higher inclinations/latitudes. Most data come from basins and grabens resulting from a late Variscan distension. Local rotations have occurred in relation to detachments and transfer-faulting during and after the Autunian. This solution was proposed to explain the deviations in declination between and within the Lodève and Saint-Affrique basins in southern France (Cogné *et al.* 1990; Diego-Orozco & Henry 1992). Krs (1968) observed that, in the Bohemian Massif, the Late Carboniferous directions are more scattered than the Permian ones, and attributed the scatter in declination to Late Variscan tectonics, i.e. to the Asturian phase.

(2) *Global drift.* In order to smooth the effect of local rotations, mean poles for given periods (Fig. 6; Table 7) were calculated. The APWP defined with these mean poles and with the few reliable poles available for the Triassic still shows a cusp which has to be interpreted in terms of global tectonics. The Late Autunian–Saxonian hair-pin implies an inversion of the sense of rotation of the European plate relative to the meridian. After the latest Visean–Namurian, the European plate and the accreted Variscides were rotating clockwise (Edel 1987). This motion stopped in the Late Autunian–Early Saxonian and, up to the Late Trias, the motion became a counterclockwise rotation coupled with a northward drift.

Table 7. Mean virtual geomagnetic poles calculated from Table 6.

Age	Ma	Lat	Long	N	k	α95	Lab
Westph.-Steph.	303	32	153	8	664	2	WS
Autunian	292	41	169	21	170	2	A
Autun.-Saxon.	272	44	177	7	81	7	AS
Saxon.-Thur.	258	49	167	13	102	4	ST

CONCLUSIONS: GEOTECTONIC IMPLICATIONS

This palaeomagnetic investigation demonstrates that at least 90 per cent of the volcanic rocks from the northern Vosges are overprinted. The series was emplaced around the Stephanian–Autunian boundary, i.e. around 298 Ma, while overprinting lasted up to the Late Permian. Maximum overprinting occurred in Saxonian–Thuringian times, connected with hydrothermal alteration of titano-magnetites into haematite and crystallization of secondary haematite in cracks and clivages of altered biotites and amphiboles. This phenomenon was also observed in older, Devonian–Dinantian units from the Vosges, Pfalz, Odenwald and Spessart. The poles corresponding to high-temperature overprints fit fairly well with the Thuringian poles of southern France (Edel & Wickert 1991; Edel & Lewandowski 1993). On the other hand, in granitoids from the southern Bohemian Massif, this late overprinting is related to crystallization of low-temperature minerals interpreted as sulfides (Reisinger *et al.* 1994).

As the volcanics have recorded the magnetic field during a large part of the Permian, the geotectonic evolution of the Vosges during this period can be inferred. Comparisons with other results from the European Variscides show that the evolution of the Vosges is valid for the whole of Variscan Europe. The following motions can be deduced from the APWP.

(1) *A clockwise rotation during the Late Carboniferous–Early Permian* which is correlated with dextral wrench-faulting in the Variscan belt (Arthaud & Matte 1977) and with the closure of the Theic ocean between Laurentia and Africa–Variscan Europe; this rotation was initiated in Late Visean–Namurian times when the Hercynides were colliding in the north-east (Edel 1987; Edel & Wickert 1991; Edel & Lewandowski 1993).

(2) *End of the rotation in the mid-Permian (Late Autunian–Early Saxonian).* This event correlates with suturing of the southern part of the Theic ocean and with the latest Alleghanian deformations in the Appalachians, i.e.

in the south-western part of the Hercynides (Secor *et al.* 1986; Piqué & Skehan 1992). In the Latest Autunian, compressive tectonics affect the Carboniferous–Early Permian basins of Iberia, Massif Central, Saar-Lorraine and mark the end of the Variscan tightening (Prost & Becq-Giraudon 1989; Broutin *et al.* 1994). This blocking situation implies that the Gondwana and Laurussia plates slowed down and changed their directions of drifting.

(3) *A counterclockwise rotation and a northward drift in Saxonian–Thuringian times*, which lasted until the Late Triassic. From the palaeomagnetic point of view, this period is characterized by large-scale hydrothermal alteration of iron-oxides and mafic minerals leading to magnetic overprinting. In the central part of the Variscides the affected rocks mostly show a reddish pigmentation resulting from haematization. The weathering process probably results from a combination of (a) the distensive regime subsequent to the end of the Variscan collision, (b) massive erosion of the Variscan mountain belt and (c) climatic changes due to the drift from equatorial to tropical latitudes of the mega plate Pangea.

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